ETH zürich

Mobile Computing

Emerging Technologies
5G Cellular Networks

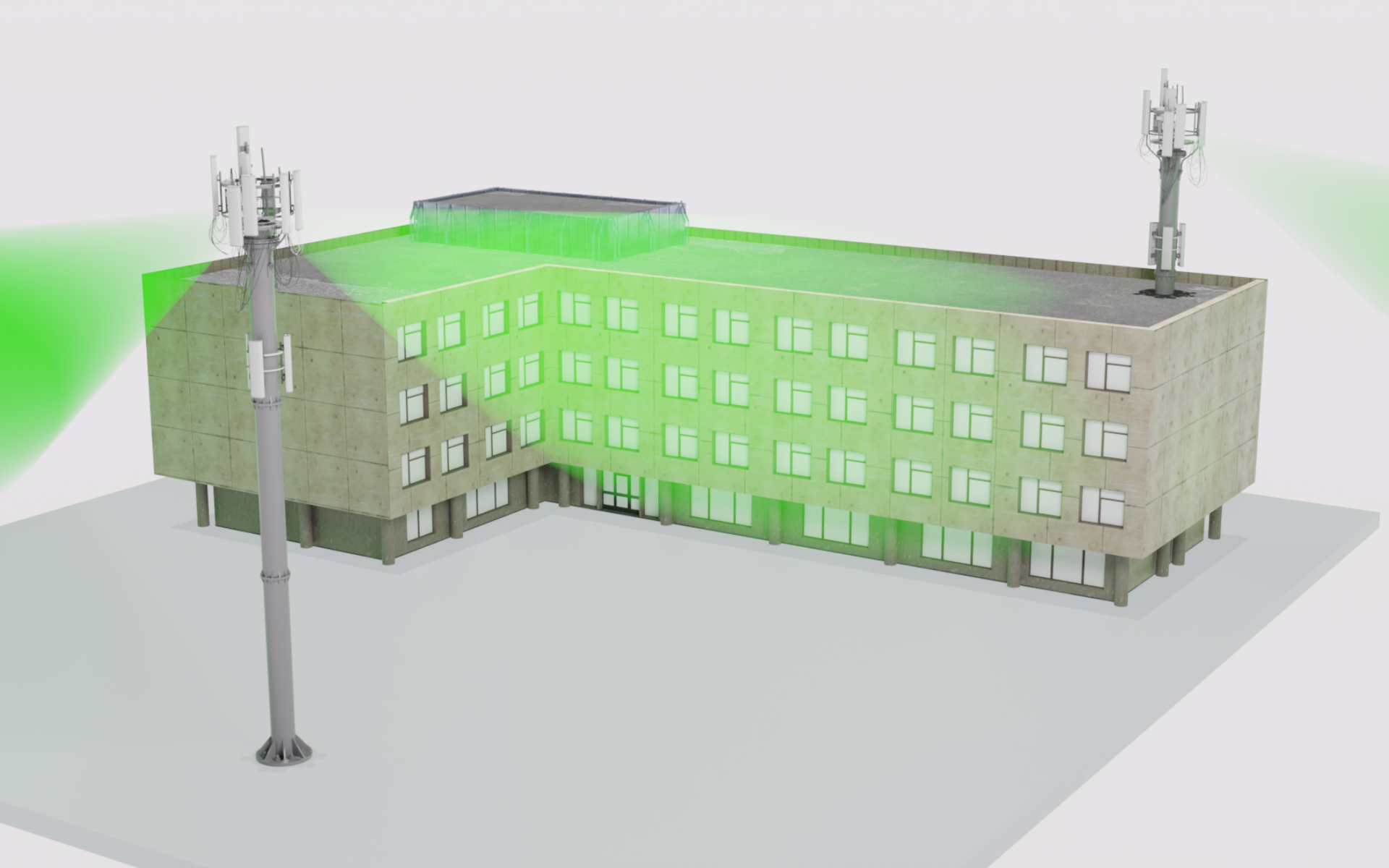
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Course Outline

Introduction

1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament



Sources

- This document contains illustrations and charts from Swisscom, Ericsson, Huawei, Qualcomm, Standards Body 3GPP



History of 3GPP Cellular Standards

- 2G = GSM (Mobile Voice Calls)
- 3G = UMTS (Mobile Internet)
- 4G = Long Term Evolution (4G LTE)
  - All IP Core Network
- 4G LTE Advanced
  - Higher data rate with reduced latency
  - increased network capacity and coverage
- 5G improves all of this
  - Efficient & flexible spectrum
- 6G = under development, “AI-Native”

Exercise: 5G Mobile Communication

- Open in your course folder:
  02_Notebooks/03_Mobile_Systems/
  03.1_5G_Mobile_Communication.ipynb
  - Run the cells and answer the questions.
  - Save the completed notebook and upload it to Moodle.





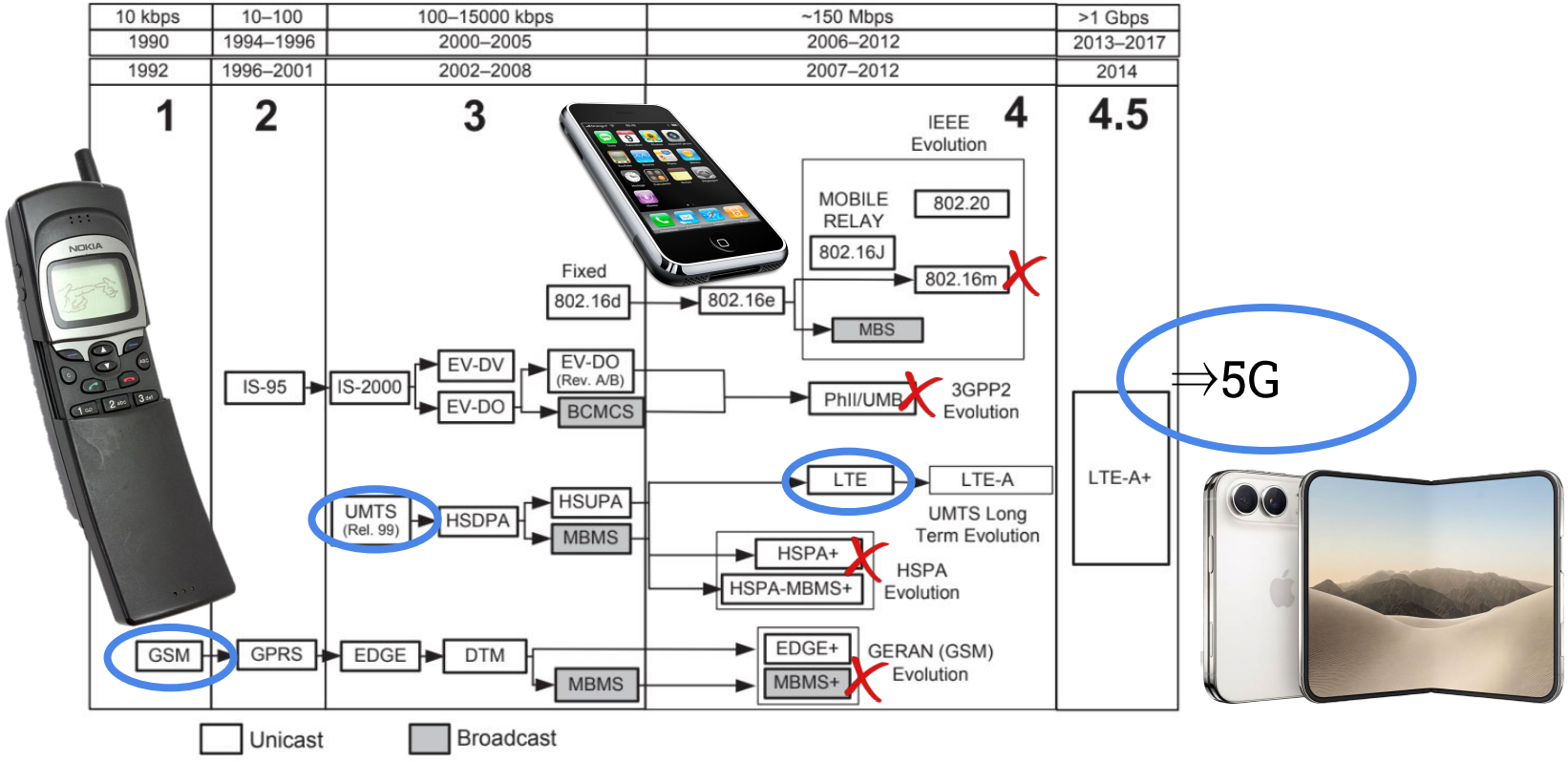


operator co-location
(tower in FL, USA)


ROSS
SKYTRAK



History of Cellular Standards

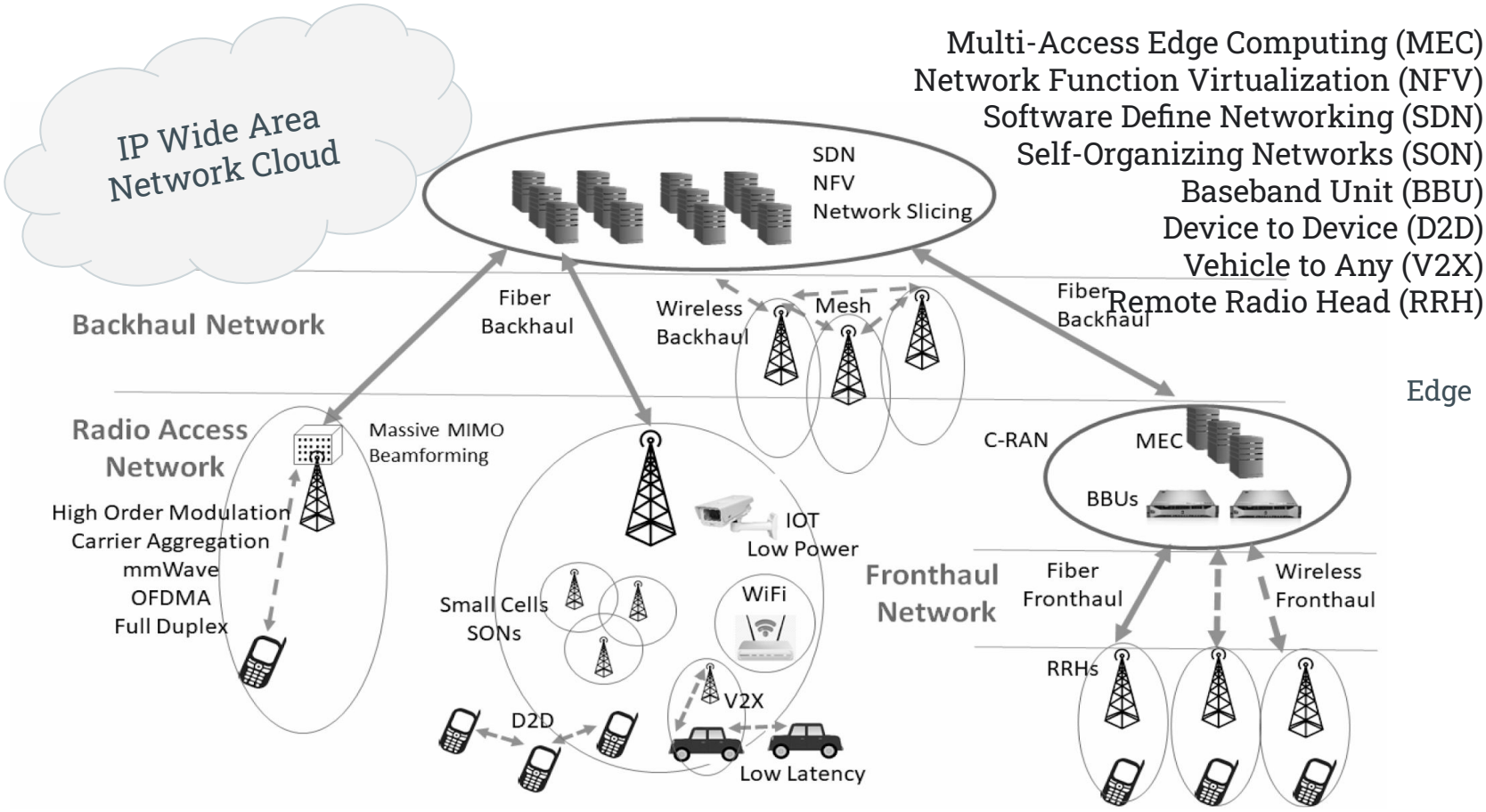


Essentials of LTE and LTE-A. Amitabha Ghosh, Rapeepat Ratasuk. Book DOI: dx.doi.org/10.1017/CBO9780511997082

Cellular Network Architecture

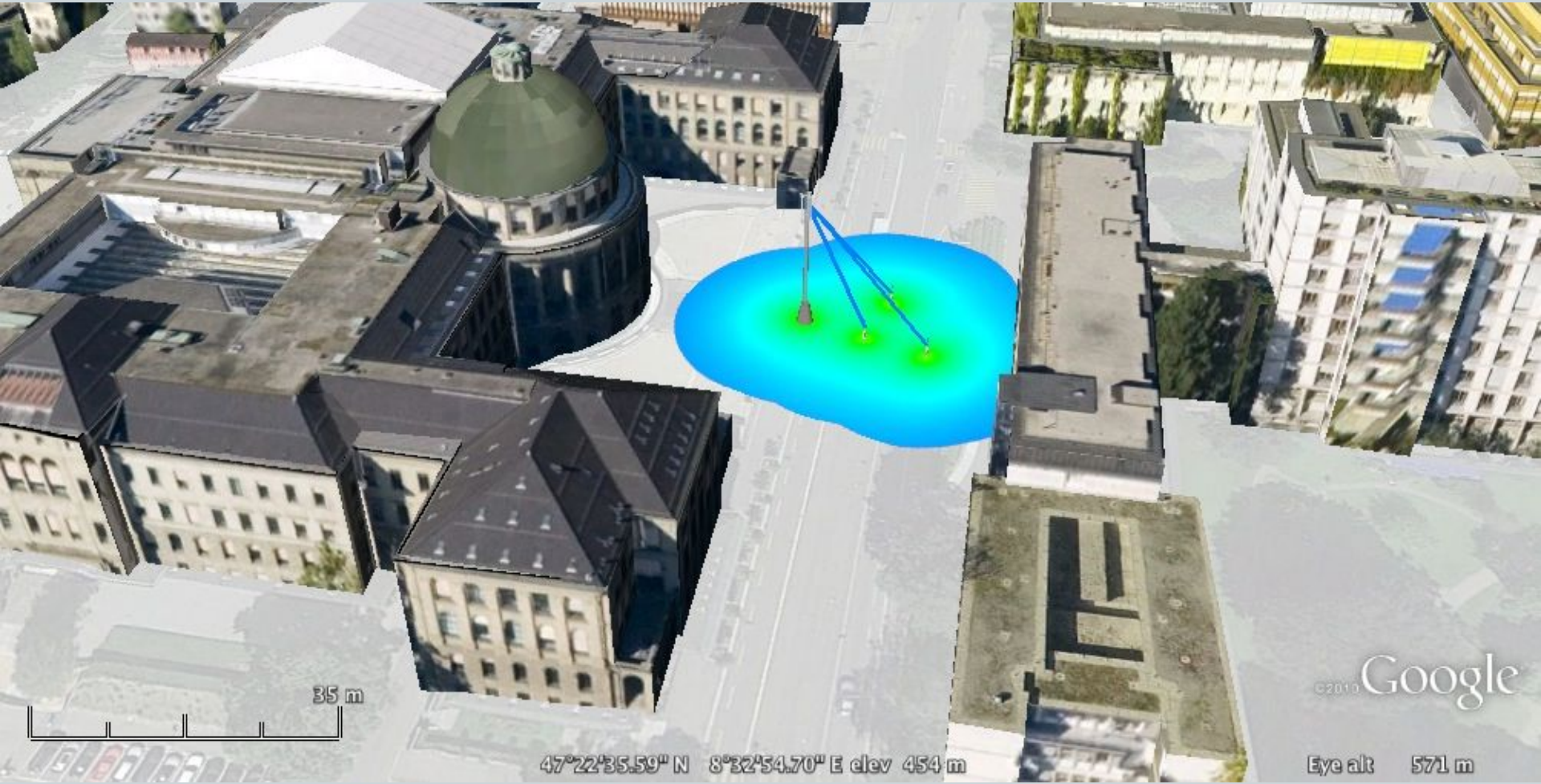
- Multiple low power transmitters, 10 Watt or less
- Coverage area is divided into cells
  - Each with own antenna
  - Each with own range of frequencies
  - Served by base station: Transceiver, control unit
- Adjacent cells use different frequencies to reduce interference

Infrastructure



Multi-Access Edge Computing (MEC)
Network Function Virtualization (NFV)
Software Define Networking (SDN)
Self-Organizing Networks (SON)
Baseband Unit (BBU)
Device to Device (D2D)
Vehicle to Any (V2X)
Remote Radio Head (RRH)

Radio Cell Coverage



Coverage Limit



Capacity Limit



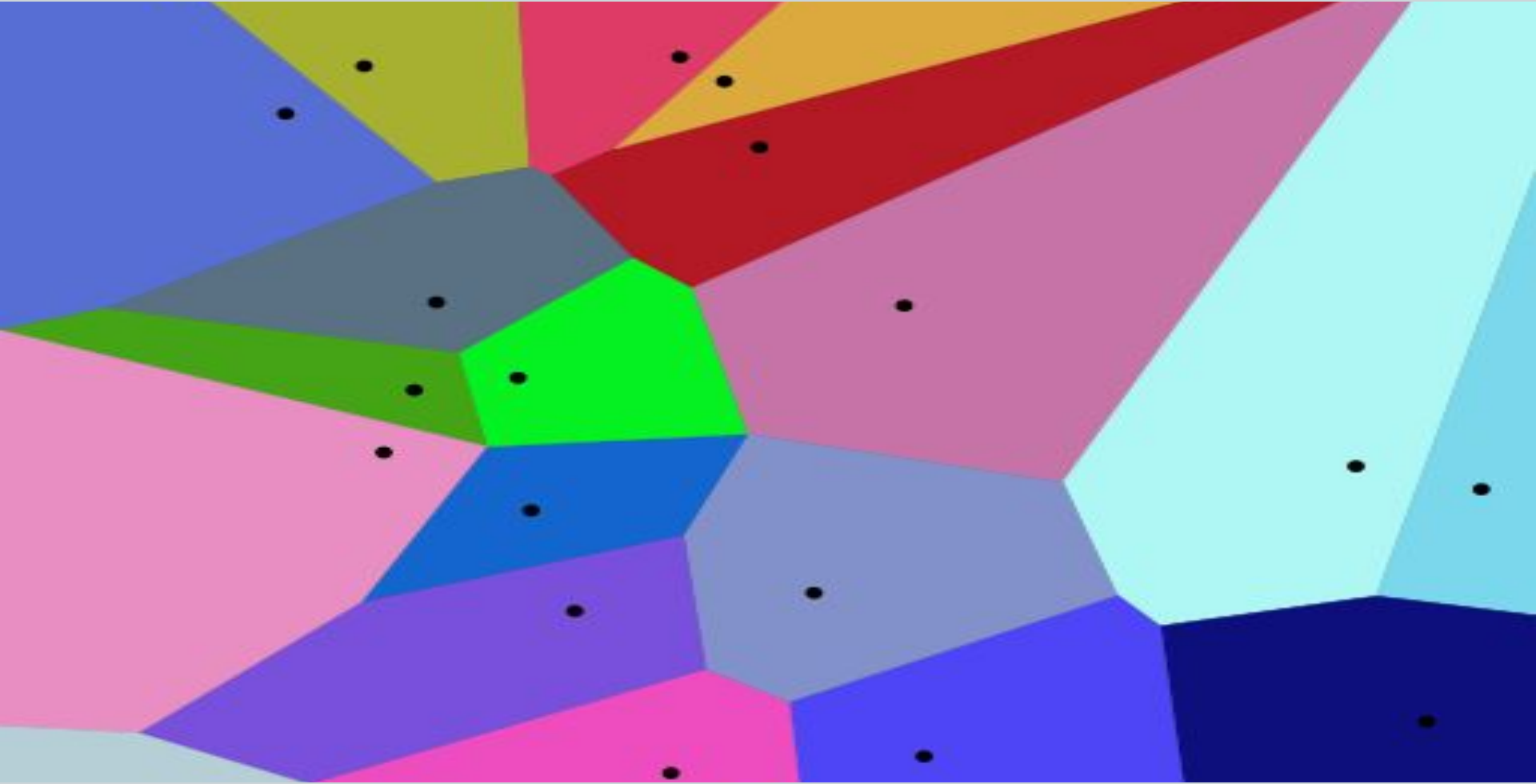
Cellular Network Planning

$$d_{\text{coverage}} = f(P_{\text{TX}}, P_{\text{RX, min}}, L_{\text{path}})$$

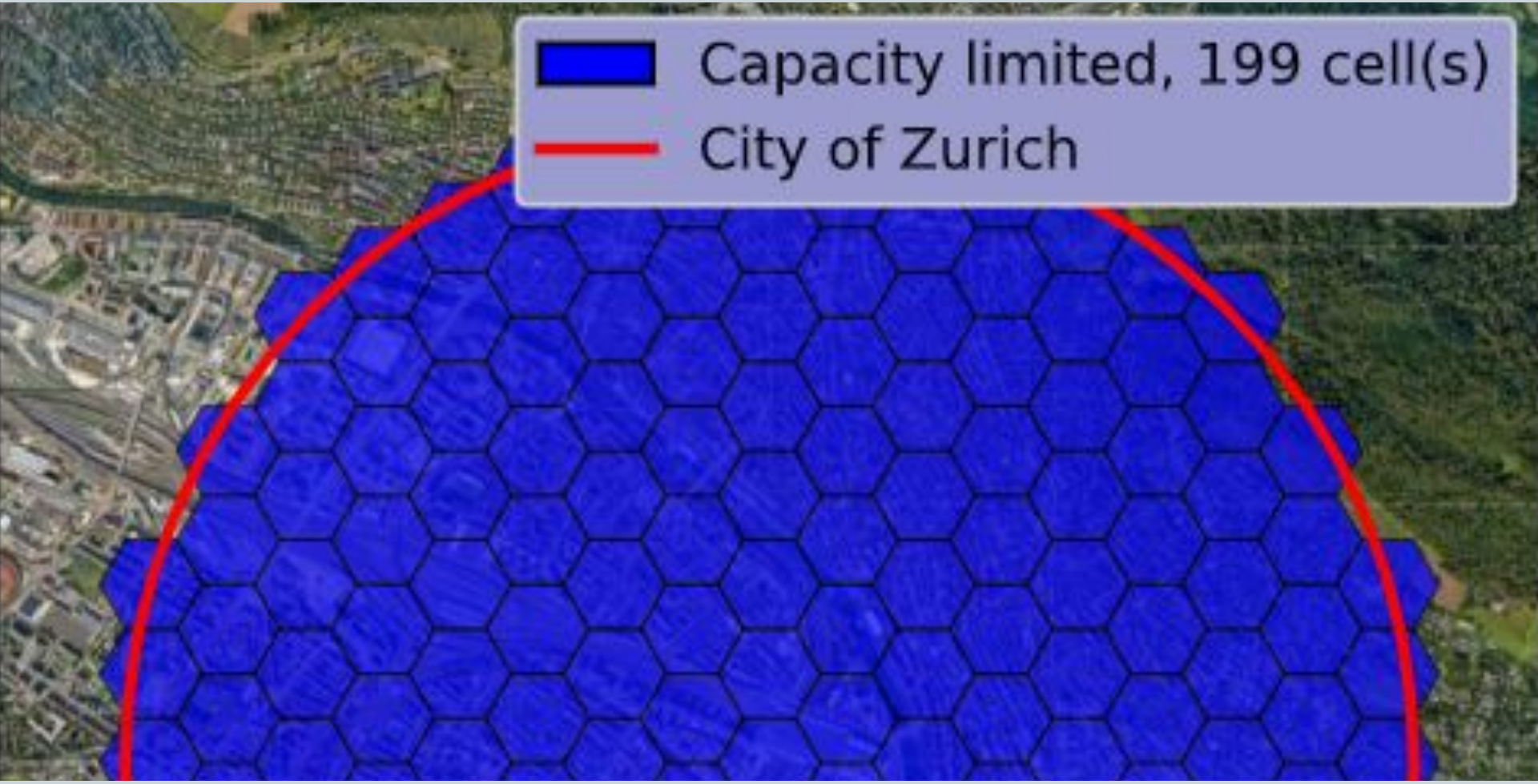
$$d_{\text{capacity}} \propto \sqrt{\frac{C}{D_{\text{cell}}}}$$

$$d_{\text{cell}} = \min(d_{\text{coverage}}, d_{\text{capacity}})$$

Realistic Cell Shapes



Network Planning uses Hexagonal Grid

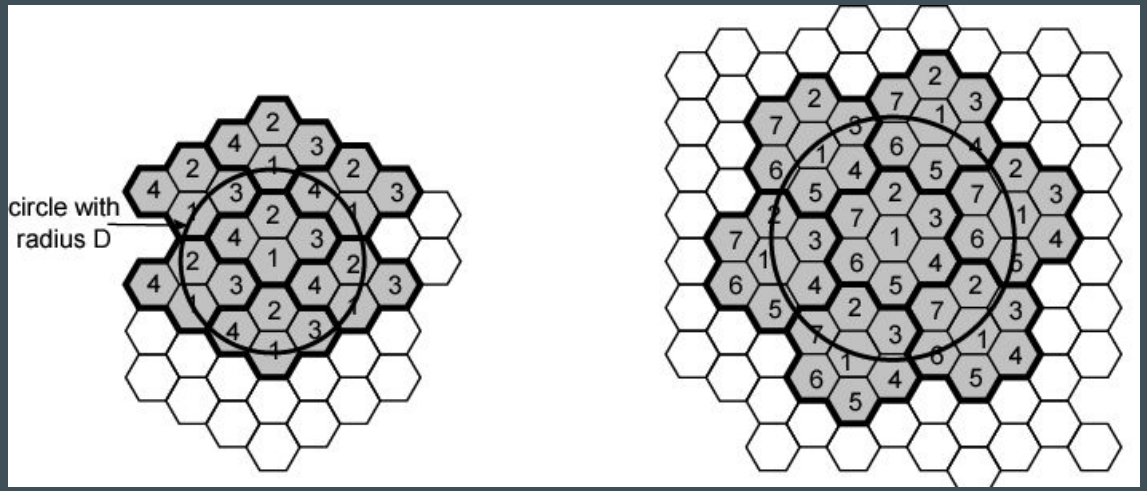


Reuse of Radio Resources (Frequency Channels)

- Use the same frequency channel at different locations
  - Communication in one cell on a given frequency channel
  - Reuse of this frequency channel in a nearby cell
- Typically, 10 - 50 frequency channels per base station
  - For example: N = number of cells in one cluster (for example clusters of 3 frequencies for reuse)
  - K = total number of frequencies used in the network (nationwide) -> Each cell operates with K/N frequency channels

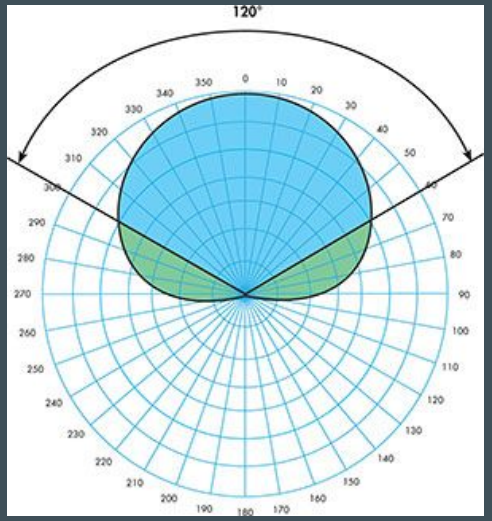
Reuse Pattern, Clusters

- Larger clusters improve signal quality S/(N+I)
- Less channels are available per cell with larger clusters

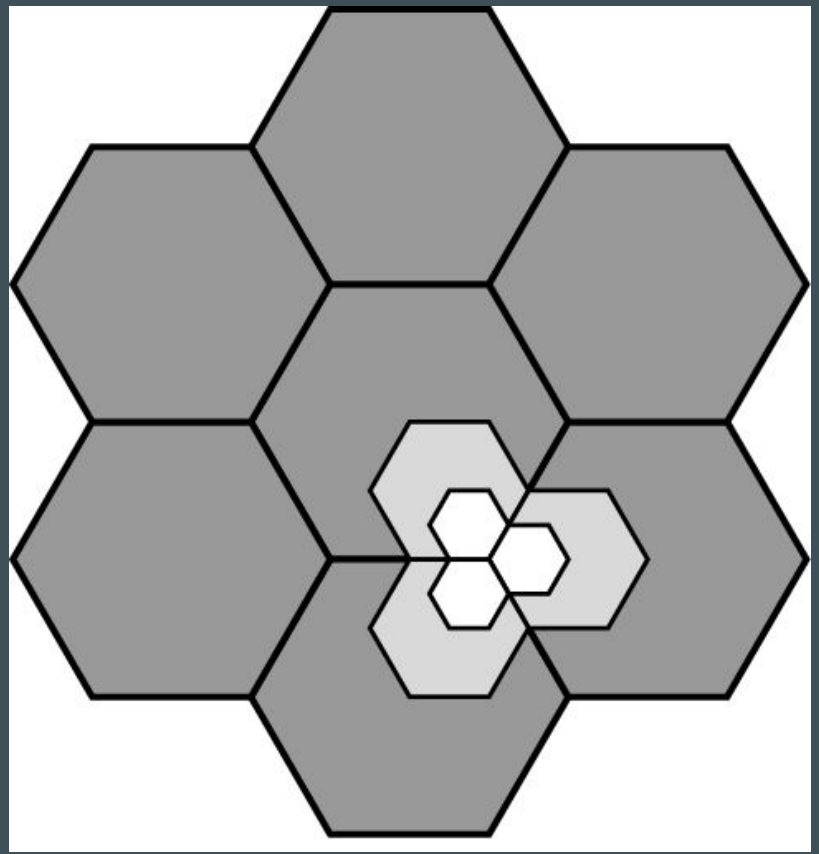


Increasing the Network Capacity ($b/s/m^2$)

- Add frequency channels, new radio spectrum
- Increase number of cells per area, more antennas
  - smaller cell sizes
  - microcells, one antenna site per room
- Sectorize a cell
  - 3 – 6 sectors per cell
  - Sectorized (directional) antennas
- Dynamic frequency borrowing
  - Take frequencies from other cells
  - Also from other network operators



Cell Splitting with Micro Cells





In-House Micro Cell

Exercise: Cellular Networking

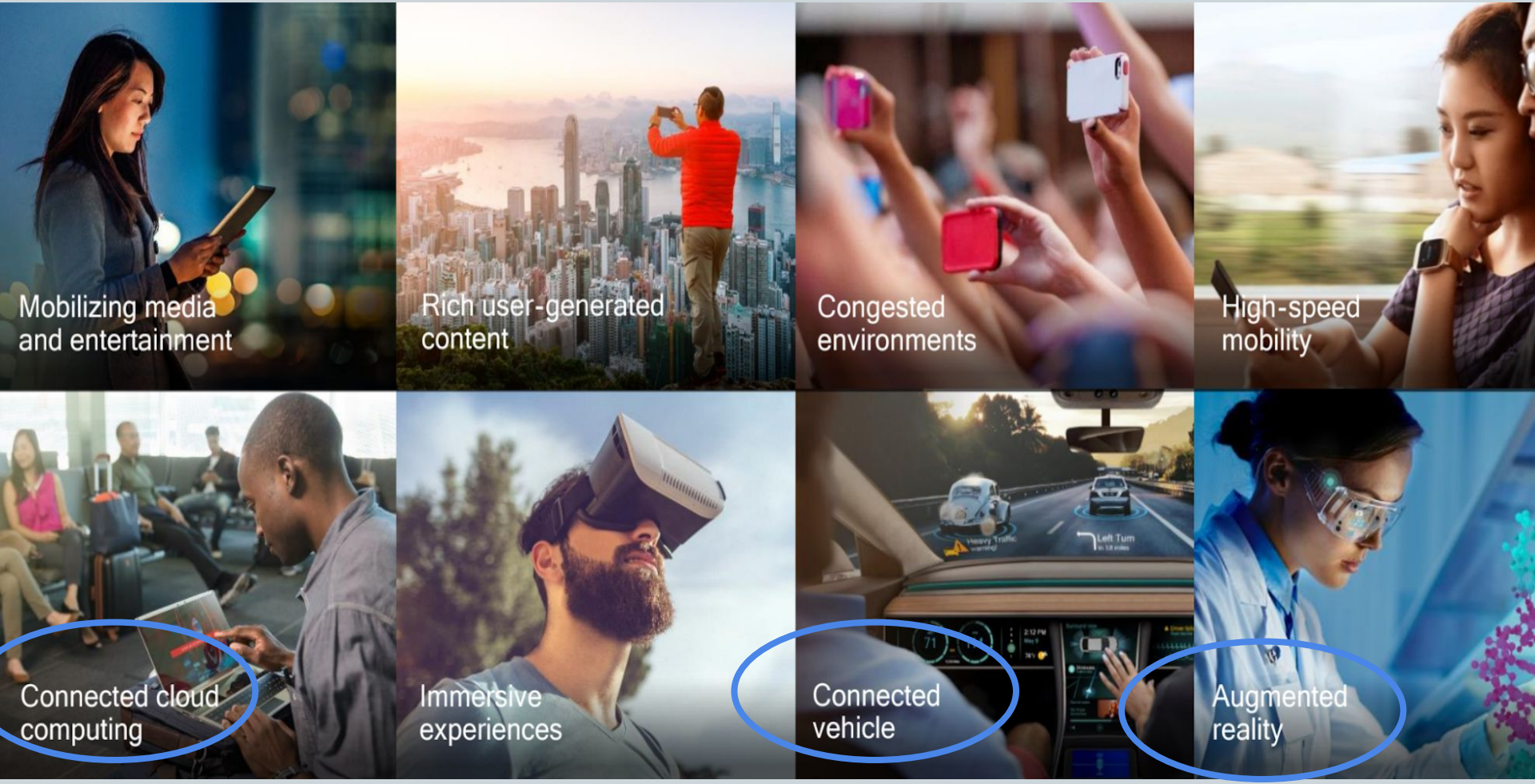
- Open in your course folder:
  02_Notebooks/03_Mobile_Systems/
  03.2_Cellular_Networking.ipynb
  - Run the cells and answer the questions.
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5G

5G Applications

Market	Mobile Application	QoS Requirement
Mobile Broadband	fast broadband, cloud-based storage	hoher thrp (10 Gb/s), low latency
VR/AR/Hologram	live broadcast, design, games	high thrp, low latency
Smart City	IoT connected devices, cloud computing, traffic mgmt.	low latency
Automotive	autonomous driving, massive sensor, edge computing	low latency
Smart Manufacturing	edge computing, logistics, cameras	high thrp, low latency
Drones	network planning	low latency
Wearables	body cameras, personal safety monitoring	low latency
eHealth	remote patient monitoring, diagnostics	low latency
Retail / Payment	interactive displays, e-kiosks	high thrp
Energy	intelligent power distribution, smart meters	high thrp

5G Applications (QUALCOMM)

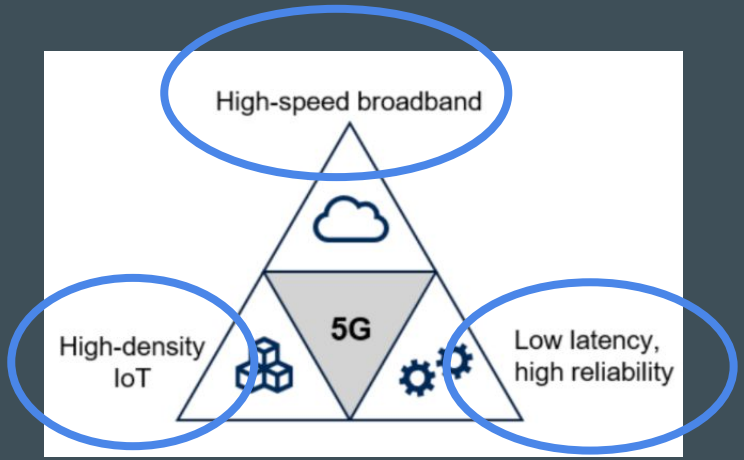


5G Applications (QUALCOMM)



2018, Gartner: Three Key Drivers for 5G

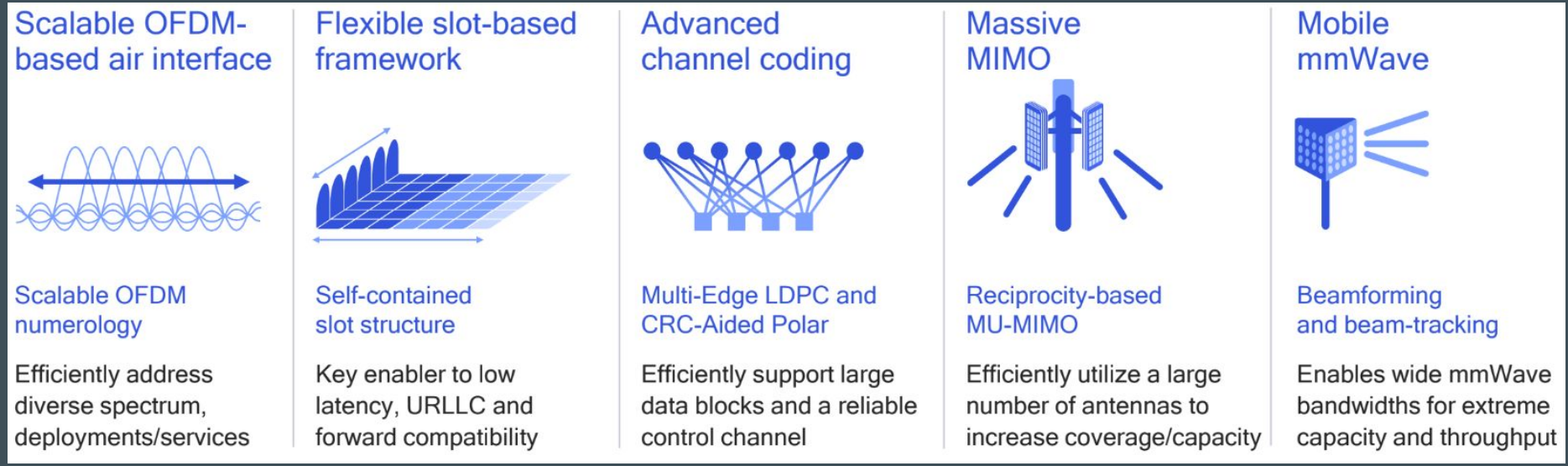
- eMBB: enhanced Mobile Broadband
- URLLC: Ultra Reliable Low Latency Comm.
- mMTC: massive Machine Type Communications



3GPP Releases

Jahr	Release	Main Objectives
2018	Rel-15	5G Basic: eMBB, Basic URLLC
2019	Rel-16	5G Evolution: V2X, NR-U, IIoT/TSN, IAB, Positioning
2021	Rel-17	5G Evolution: eMBB, URLLC, mMTC Features
2023	Rel-18	5G Advanced: RedCap, Non Terrestrial Networks
2025	Rel-19	5G Advanced
2026	Rel-20	5G Advanced
2028	Rel-21	6G Basic: Terahertz Communication, Intelligent Surfaces

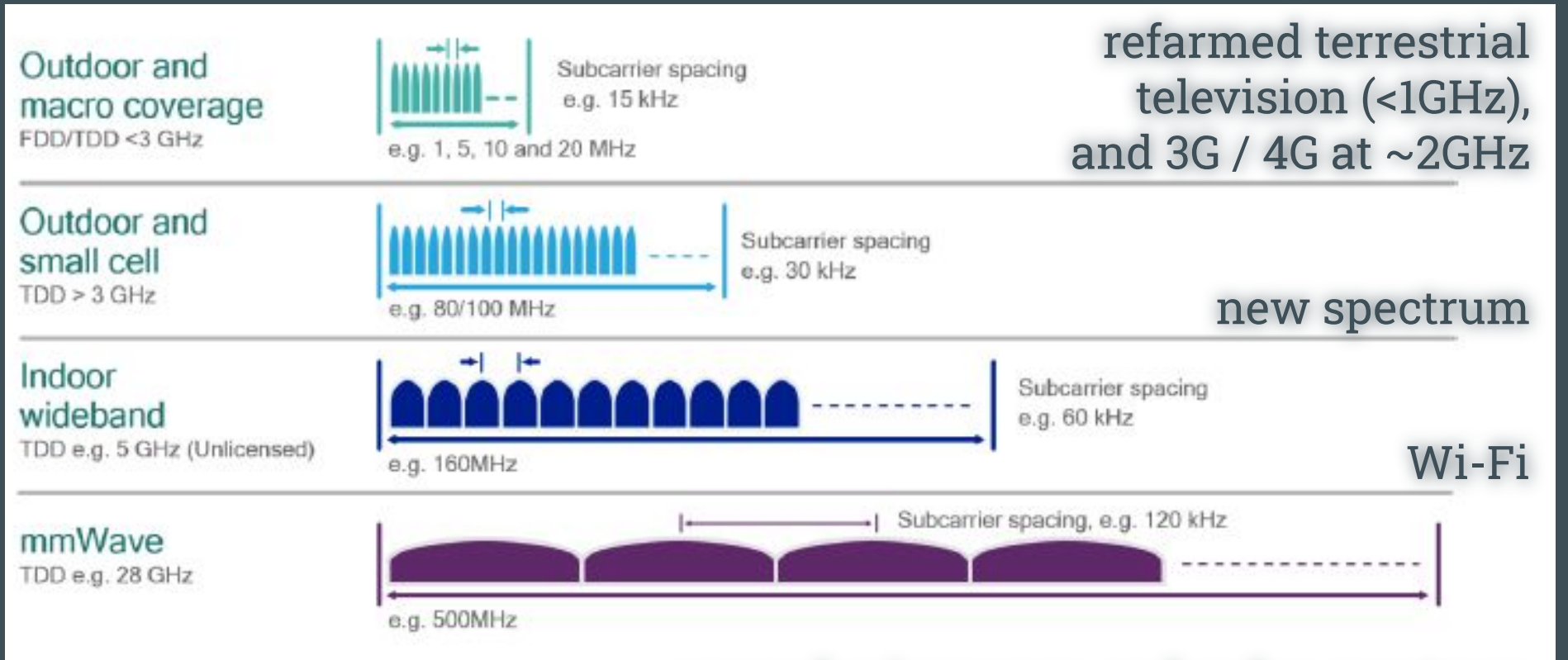
5G “New Radio” Air Interface



Source: Qualcomm

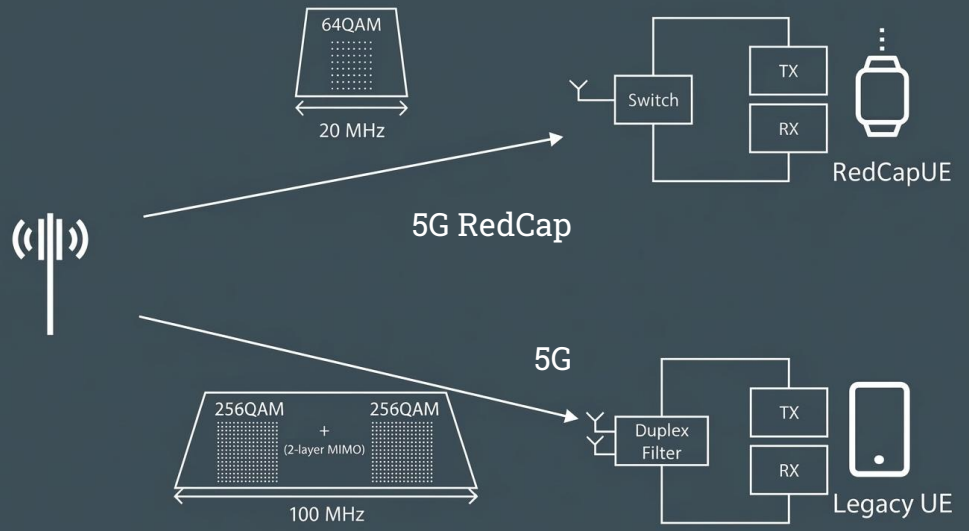
- Pretty much the same as Wi-Fi, but now using licensed AND unlicensed radio spectrum

5G “New Radio” Air Interface



5G “New Radio” Air Interface - RedCap

- Reduced Capability (Source: Ericsson)
- Simplifies a lot, for example:
  - Reduced Bandwidth, 20 MHz (example)
  - Only one Transceiver, Half Duplex FDD
  - 2x2 MIMO
  - 2 … 6 GHz
- Similar to Wi-Fi 4
  IEEE 802.11n, 2003

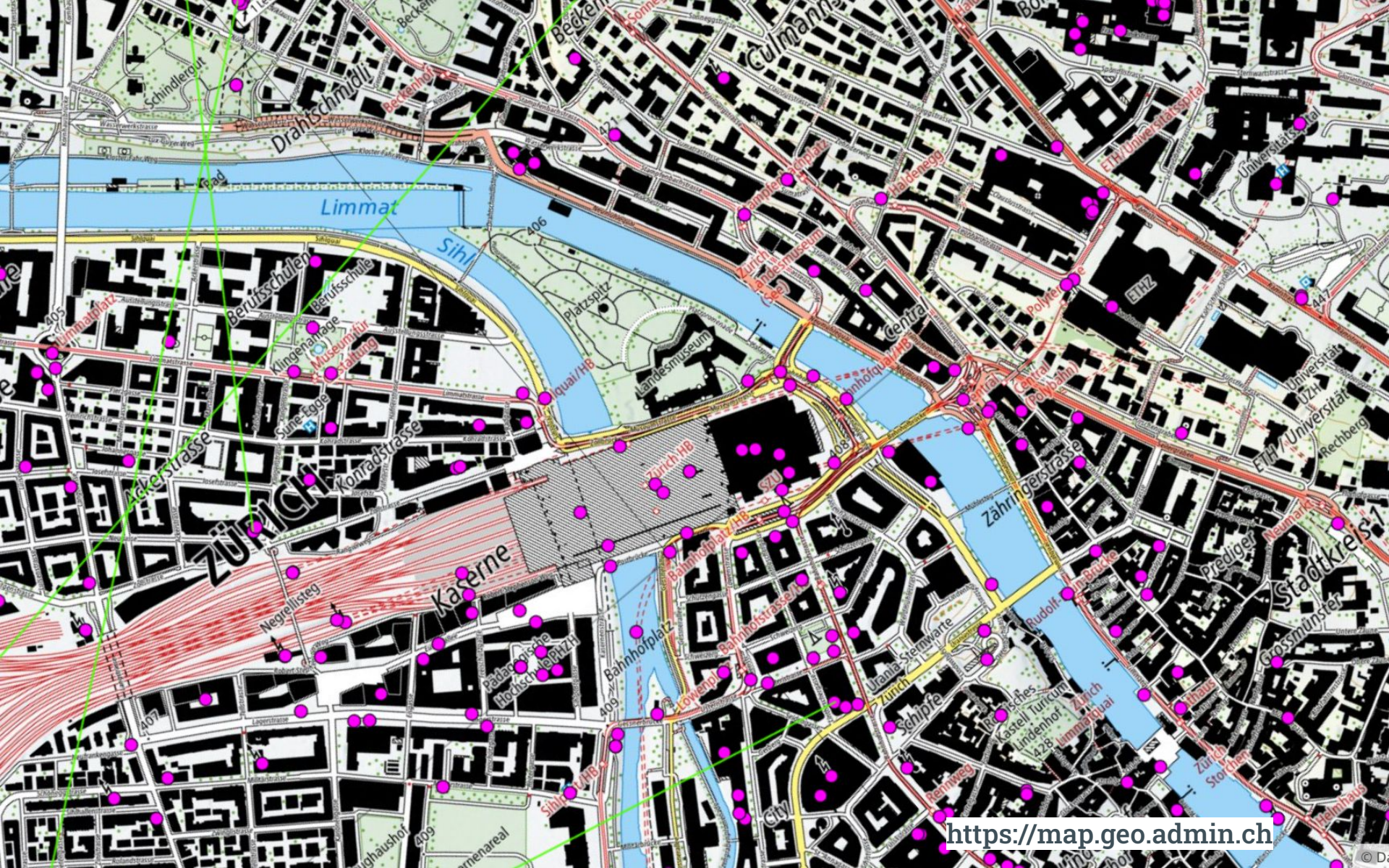


Exercise: 5G RedCap for the Future IoT

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.9_5G_RedCap_for_Future_IoT.ipynb
  - Run the cells and answer the questions.
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NFV and SDN

- Network Functionality Virtualization (NFV)
  - Defines network elements such as IP packet routers
- Software Defined Networking (SDN)
  - Manages large parts of the network
- NFV and SDN separate infrastructure from operation
- Low entry level for newcomers
- More MVNOs (Mobile Virtual Network Operator)


Limmat
Sihl
ZÜRICH
Kaserne
Zähringerstrasse
Bahnhofplatz
Bahnhofstrasse
Rudolf-Brun-Brücke
Stadtkreis 1
Grossmünster
Predigerkirche
Urania-Sternwarte
Schipfe
Lindenhof
Platzspitz
Landesmuseum
Zürich HB
Hauptbahnhof
Polybahn
Universitätsspital
Universität
ETH/Universitätsspital
Rechberg
Neumarkt
Limmatplatz
Berufsschulen
Museumstrasse
Konradstrasse
Sihlquai/HB
Bahnhofplatz/HB
Bahnhofquai/HB
Pädagogische Hochschule PHZH
Negrellisteg
Lagerstrasse
Sihlquai/HB
Reitschule
Kasteli Turicum
Limmatquai
Reimweg
Zürich Storchen
Helmhaus
Becken
Gulmann
Drahtschmidli
Schindlergut
Haldeneg
Zürich Hauptbahnhof
Zürich Landesmuseum
Zürich Polybahn
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Zürich Sihlquai/HB
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Zürich Storchen
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Zürich Predigerkirche
Zürich Neumarkt
Zürich Rechberg
Zürich Universität
Zürich Universitätsspital
Zürich Haldeneg
Zürich Becken
Zürich Gulmann
Zürich Drahtschmidli
Zürich Schindlergut
Zürich Limmatplatz
Zürich Berufsschulen
Zürich Museumstrasse
Zürich Konradstrasse
Zürich Sihlquai/HB
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Zürich Bahnhofquai/HB
Zürich Pädagogische Hochschule PHZH
Zürich Negrellisteg
Zürich Lagerstrasse
Zürich Sihlquai/HB
Zürich Reitschule
Zürich Kasteli Turicum
Zürich Limmatquai
Zürich Reimweg

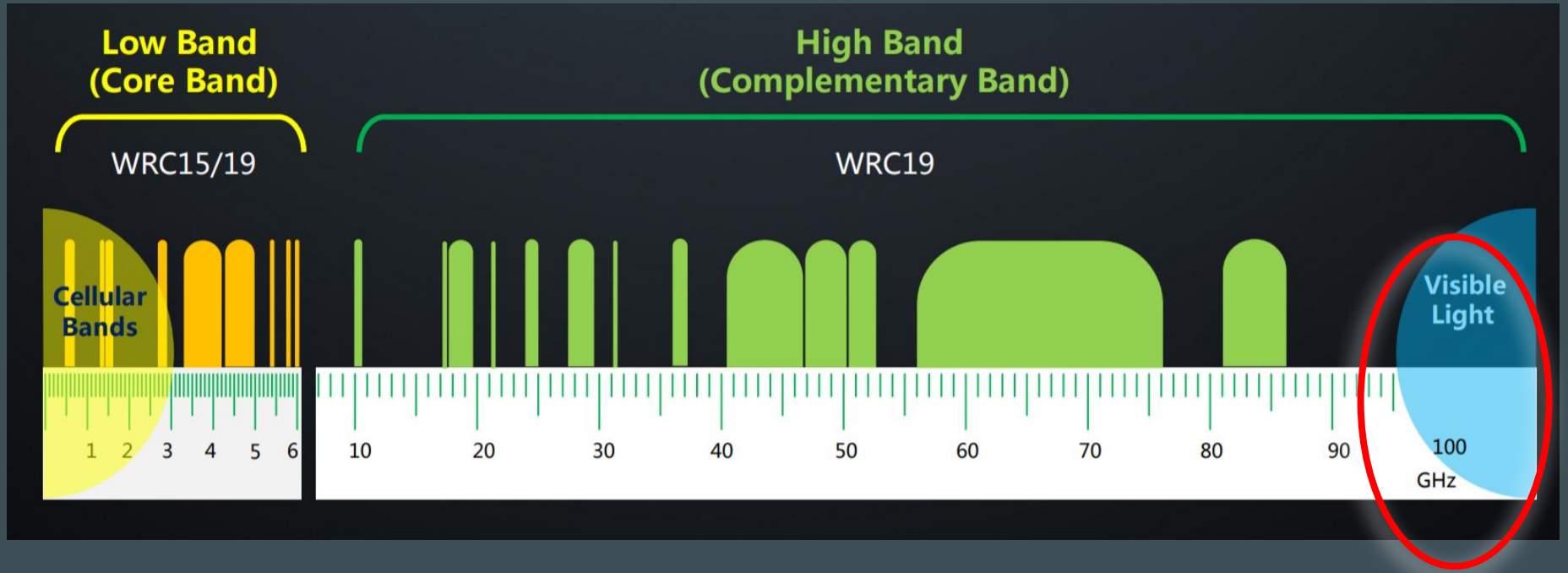
6G

Reconfigurable Intelligent Surfaces



6G and Visible Light Spectrum

- VLC might be considered in the future (6G)
  - WRC = World Radiocommunication Conferences



3GPP Release 18 (2024)

- Called “5G-Advanced”, improved generic enablers
  - Network Slicing, Edge Computing, Proximity Services

https://www.3gpp.org/ftp/Inbox/Marcoms/3GPP_Poster%20v2.pdf

